BF
Autopilot Connectors

BF
Interface MCU

BF
Power System

BF
MARS Nav

BF
BlackFin Unit

MH MH MH
MH1 MH2 MH3
GND GND

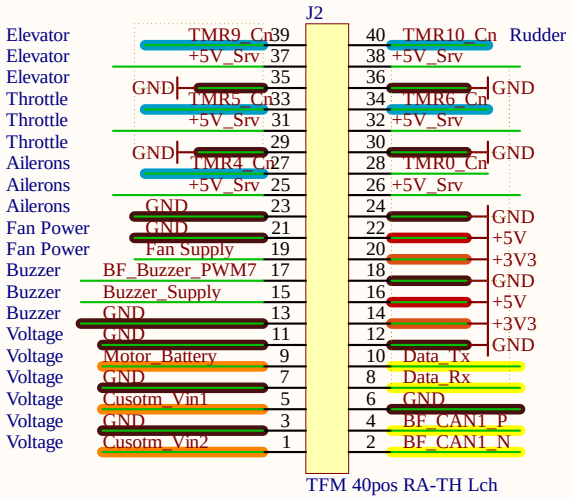
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MARS Robotics
Jordan - Irbid
Fawzi AL Mulqi Street
P.O Box 2233
Tel:+962 2 7102026

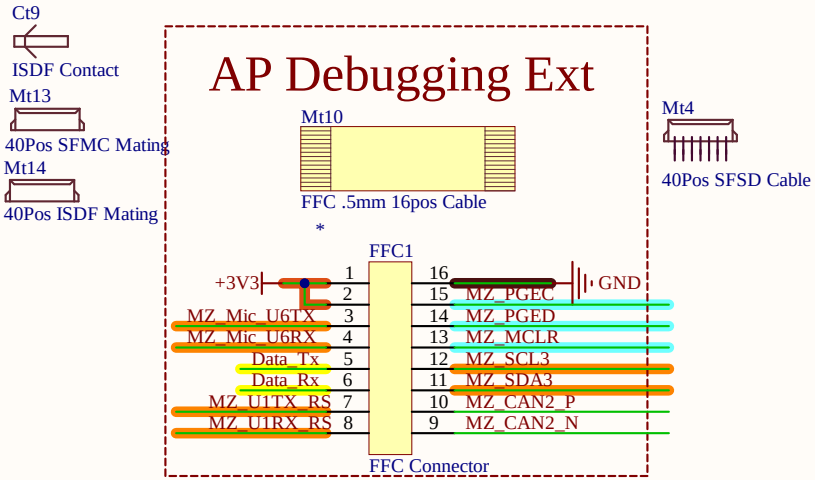
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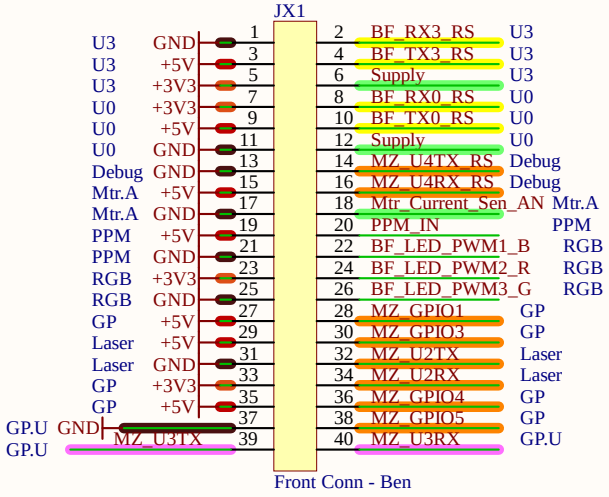
AP Connector

Front Connector

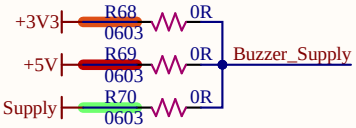


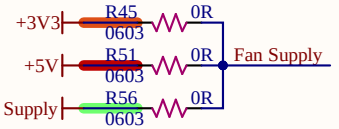
AP Debugging Ext





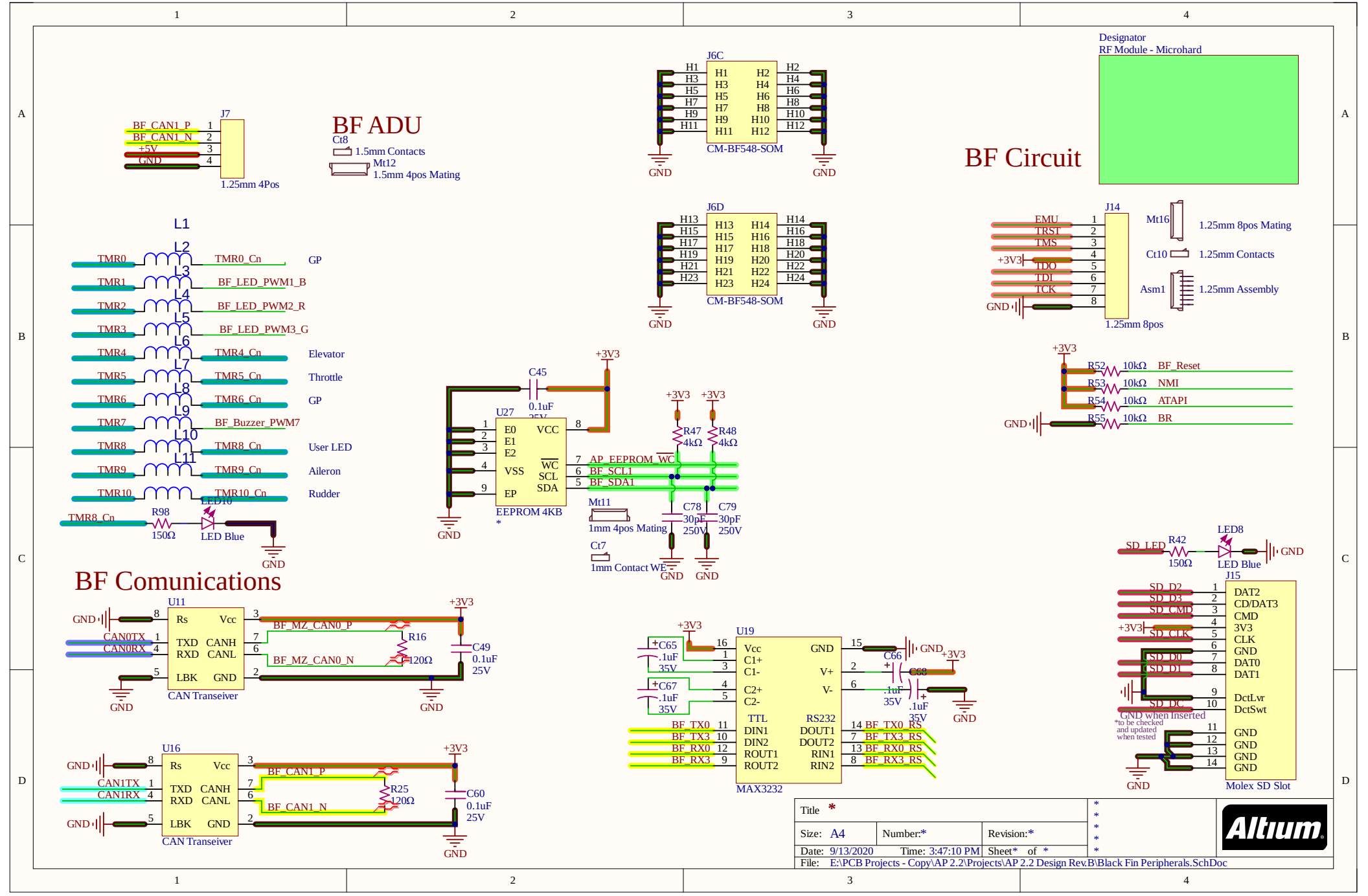
GPIO1(D4): Output Compare[5/8]/Input Capture[2/5/9]/INT2
GPIO2(B2): Output Compare[2/1/9]/Input Capture[1/6]/ INT1/AN2
GPIO3(B8): Output Compare[5/8]/Input Capture[2/5/9]/INT2/AN48
GPIO4(E3): Output Compare[5/8]Input Capture[2/5/9]/INT2
GPIO5(E5): Output Compare[4/7]/Input Capture[4/8]/INT4/AN17

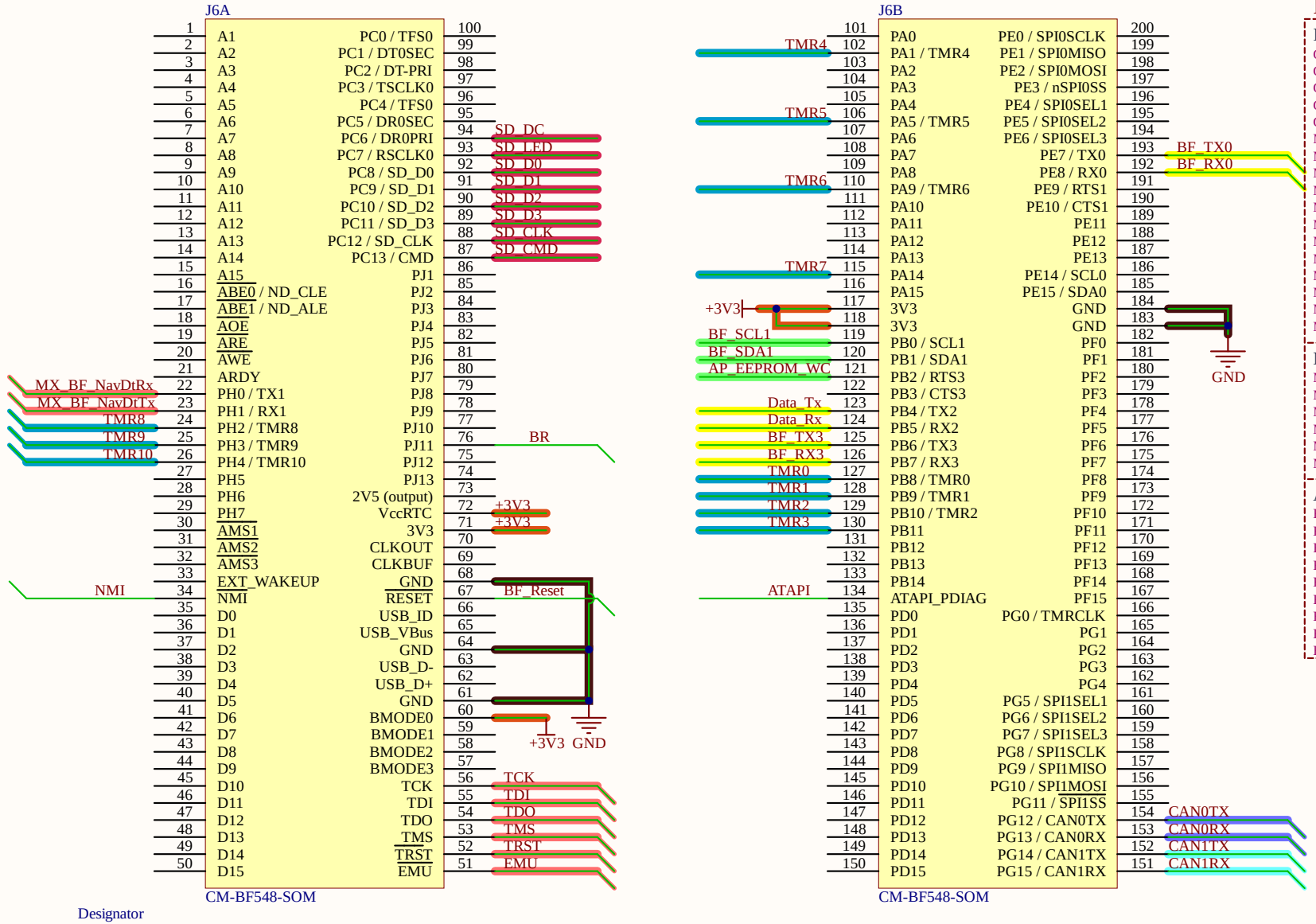


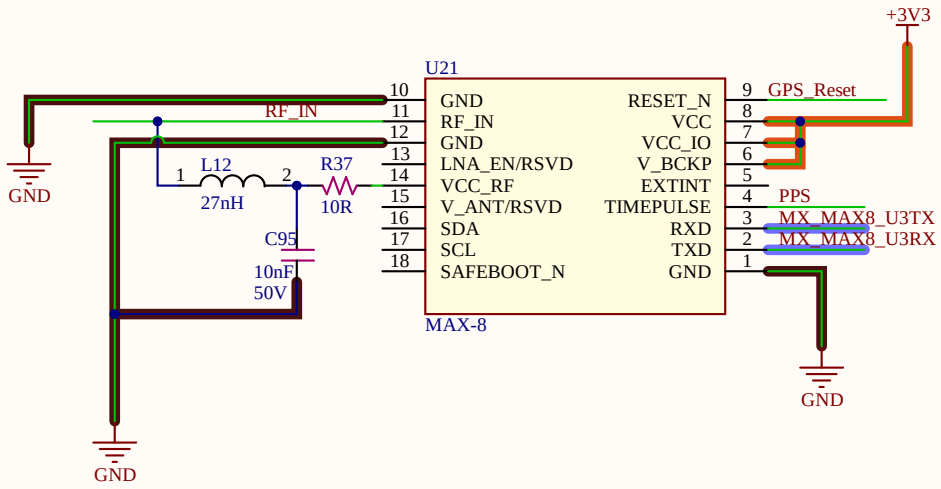


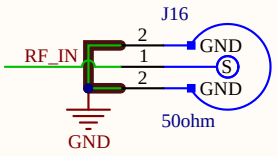
BF_LED_PWM3_G — BF_LED_PWM3_G
BF_LED_PWM2_R — BF_LED_PWM2_R
BF_LED_PWM1_B — BF_LED_PWM1_B

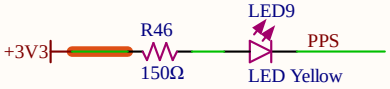
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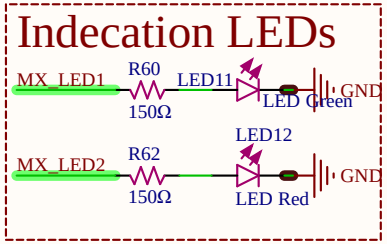

BF ADU
BF Circuit
BF Comunications
Designator
RF Module - Microhard
J7
BF_CAN1_P
BF_CAN1_N
+5V
GND
1.25mm 4Pos
Ct8
1.5mm Contacts
Mt12
1.5mm 4pos Mating
J6C
CM-BF548-SOM
J6D
CM-BF548-SOM
J14
EMU
TRST
TMS
+3V3
TDO
TDI
TCK
GND
1.25mm 8pos
Mt16
1.25mm 8pos Mating
Ct10
1.25mm Contacts
Asm1
1.25mm Assembly
L1
L2
L3
L4
L5
L6
L7
L8
L9
L10
L11
TMR0
TMR1
TMR2
TMR3
TMR4
TMR5
TMR6
TMR7
TMR8
TMR9
TMR10
TMR0_Cn
BF_LED_PWM1_B
BF_LED_PWM2_R
BF_LED_PWM3_G
TMR4_Cn
TMR5_Cn
TMR6_Cn
BF_Buzzer_PWM7
TMR8_Cn
TMR9_Cn
TMR10_Cn
GP
Elevator
Throttle
GP
User LED
Aileron
Rudder
LED10
R98
150Ω
LED Blue
C45
0.1uF
U27
EEPROM 4KB
AP_EEPROM_WC
BF_SCL1
BF_SDA1
R47
4kΩ
R48
4kΩ
Mt11
1mm 4pos Mating
C78
30pF
250V
C79
30pF
250V
Ct7
1mm Contact WE
R52
10kΩ
BF_Reset
R53
10kΩ
NMI
R54
10kΩ
ATAPI
R55
10kΩ
BR
SD_LED
R42
150Ω
LED8
LED Blue
J15
SD_D2
SD_D3
SD_CMD
SD_CLK
SD_D0
SD_D1
SD_DC
DAT2
CD/DAT3
CMD
3V3
CLK
GND
DAT0
DAT1
DctLvr
DctSwt
GND when Inserted
*to be checked and updated when tested
Molex SD Slot
U11
CAN Transeiver
CAN0TX
CAN0RX
BF_MZ_CAN0_P
BF_MZ_CAN0_N
R16
120Ω
C49
0.1uF
25V
U16
CAN Transeiver
CAN1TX
CAN1RX
BF_CAN1_P
BF_CAN1_N
R25
120Ω
C60
0.1uF
25V
U19
MAX3232
C65
.1uF
35V
C67
.1uF
35V
C66
.1uF
35V
C68
.1uF
35V
BF_TX0
BF_TX3
BF_RX0
BF_RX3
BF_TX0_RS
BF_TX3_RS
BF_RX0_RS
BF_RX3_RS
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J6A
1 A1 PC0 / TFS0 100
2 A2 PC1 / DT0SEC 99
3 A3 PC2 / DT-PRI 98
4 A4 PC3 / TSCLK0 97
5 A5 PC4 / TFS0 96
6 A6 PC5 / DR0SEC 95
7 A7 PC6 / DR0PRI 94 SD_DC
8 A8 PC7 / RSCLK0 93 SD_LED
9 A9 PC8 / SD_D0 92 SD_D0
10 A10 PC9 / SD_D1 91 SD_D1
11 A11 PC10 / SD_D2 90 SD_D2
12 A12 PC11 / SD_D3 89 SD_D3
13 A13 PC12 / SD_CLK 88 SD_CLK
14 A14 PC13 / CMD 87 SD_CMD
15 A15 PJ1 86
16 ABE0 / ND_CLE PJ2 85
17 ABE1 / ND_ALE PJ3 84
18 AOE PJ4 83
19 ARE PJ5 82
20 AWE PJ6 81
21 ARDY PJ7 80
22 MX_BF_NavDtRx PH0 / TX1 PJ8 79
23 MX_BF_NavDtTx PH1 / RX1 PJ9 78
24 TMR8 PH2 / TMR8 PJ10 77
25 TMR9 PH3 / TMR9 PJ11 76 BR
26 TMR10 PH4 / TMR10 PJ12 75
27 PH5 PJ13 74
28 PH6 2V5 (output) 73
29 PH7 VccRTC 72 +3V3
30 AMS1 3V3 71 +3V3
31 AMS2 CLKOUT 70
32 AMS3 CLKBUF 69
33 EXT_WAKEUP GND 68
34 NMI NMI RESET 67 BF_Reset
35 D0 USB_ID 66
36 D1 USB_VBus 65
37 D2 GND 64
38 D3 USB_D- 63
39 D4 USB_D+ 62
40 D5 GND 61
41 D6 BMODE0 60
42 D7 BMODE1 59
43 D8 BMODE2 58 +3V3 GND
44 D9 BMODE3 57
45 D10 TCK 56 TCK
46 D11 TDI 55 TDI
47 D12 TDO 54 TDO
48 D13 TMS 53 TMS
49 D14 TRST 52 TRST
50 D15 EMU 51 EMU
CM-BF548-SOM
J6B
101 PA0 PE0 / SPI0SCLK 200
TMR4 102 PA1 / TMR4 PE1 / SPI0MISO 199
103 PA2 PE2 / SPI0MOSI 198
104 PA3 PE3 / nSPI0SS 197
105 PA4 PE4 / SPI0SEL1 196
TMR5 106 PA5 / TMR5 PE5 / SPI0SEL2 195
107 PA6 PE6 / SPI0SEL3 194
108 PA7 PE7 / TX0 193 BF_TX0
TMR6 109 PA8 PE8 / RX0 192 BF_RX0
110 PA9 / TMR6 PE9 / RTS1 191
111 PA10 PE10 / CTS1 190
112 PA11 PE11 189
113 PA12 PE12 188
114 PA13 PE13 187
TMR7 115 PA14 PE14 / SCL0 186
116 PA15 PE15 / SDA0 185
+3V3 117 3V3 GND 184
118 3V3 GND 183 GND
BF_SCL1 119 PB0 / SCL1 PF0 182
BF_SDA1 120 PB1 / SDA1 PF1 181
AP_EEPROM_WC 121 PB2 / RTS3 PF2 180
122 PB3 / CTS3 PF3 179
Data_Tx 123 PB4 / TX2 PF4 178
Data_Rx 124 PB5 / RX2 PF5 177
BF_TX3 125 PB6 / TX3 PF6 176
BF_RX3 126 PB7 / RX3 PF7 175
TMR0 127 PB8 / TMR0 PF8 174
TMR1 128 PB9 / TMR1 PF9 173
TMR2 129 PB10 / TMR2 PF10 172
TMR3 130 PB11 PF11 171
131 PB12 PF12 170
132 PB13 PF13 169
133 PB14 PF14 168
ATAPI 134 ATAPI_PDIAG PF15 167
135 PD0 PG0 / TMRCLK 166
136 PD1 PG1 165
137 PD2 PG2 164
138 PD3 PG3 163
139 PD4 PG4 162
140 PD5 PG5 / SPI1SEL1 161
141 PD6 PG6 / SPI1SEL2 160
142 PD7 PG7 / SPI1SEL3 159
143 PD8 PG8 / SPI1SCLK 158
144 PD9 PG9 / SPI1MISO 157
145 PD10 PG10 / SPI1MOSI 156
146 PD11 PG11 / SPI1SS 155
147 PD12 PG12 / CAN0TX 154 CAN0TX
148 PD13 PG13 / CAN0RX 153 CAN0RX
149 PD14 PG14 / CAN1TX 152 CAN1TX
150 PD15 PG15 / CAN1RX 151 CAN1RX
CM-BF548-SOM
Designator
Black Fin Peripherals
Peripherals Usage
Interfacing PIC32MZ
GPIO1: General Purpose
GPIO2: General Purpose
GPIO3: General Purpose
GPIO4: General Purpose
GPIO5: General Purpose
MZ_UART1: MZ Debugging
MZ_UART2: General Purpose UART_3.3V
MZ_UART3: General Purpose UART_5V
MZ_UART4: Laser_RS232
MZ_UART5: MX Communication
MZ_UART6: Microhard Control
MZ_SPI4: General Purpose
MZ_I2C3: General Purpose
MZ_I2C4: Humidity and Tempereture Sensor
MZ_I2C3: General Purpose
MZ_CAN1: Blackfin Communication
MZ_CAN2: General Purpose
MARS Nav PIC32MX
MX_CAN1: Comunication BF
MX_UART3: GPS UART
MX_UART5: Communication MZ
MX_UART6: Communication BF
MX_SPI3: ADCs reading
MX_I2C: MARS Nav EEPROM
Autopilot Blackfin
BF_CAN0: Comunication MZ
BF_CAN1: ADU + General Purpose
BF_UART0: External UART-RS232
BF_UART1: MARS Nav Data UART
BF_UART2: Microhard Data UART
BF_UART3: External UART-RS232
BF_I2C1: AP EEPROM
BF_TMR0: RC Input
BF_TMR1-10: PWM Output
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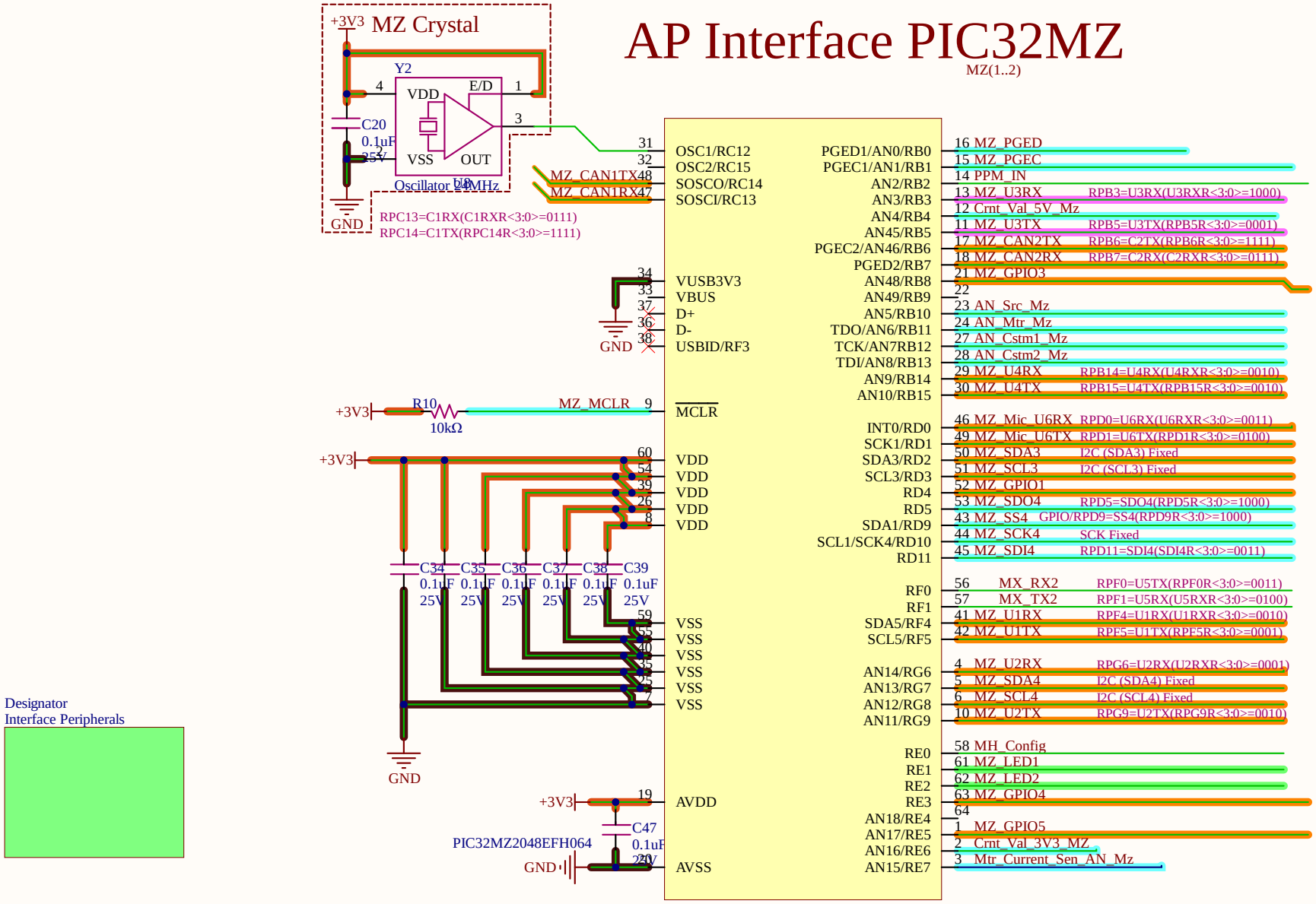




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AP Interface PIC32MZ

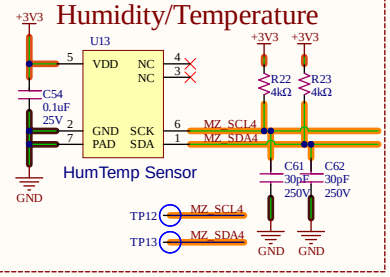
MZ(1..2)



Designator
Interface Peripherals

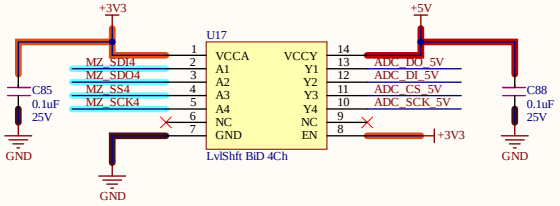
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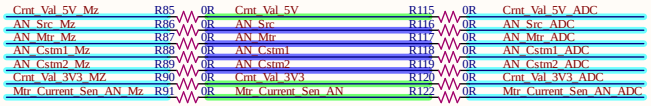
Humidity/Temperature



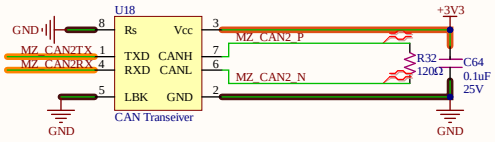
MZ_SDA3
MZ_SCL3

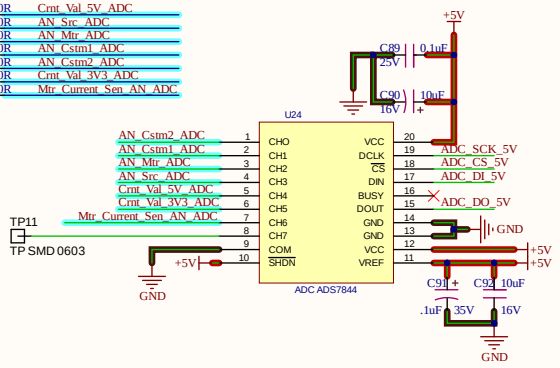
MZ_U3TX
MZ_U3RX



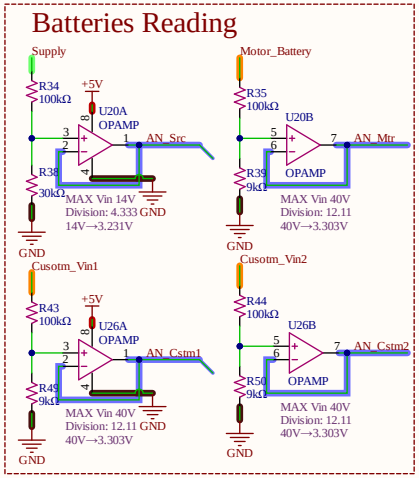


MZ Debugging

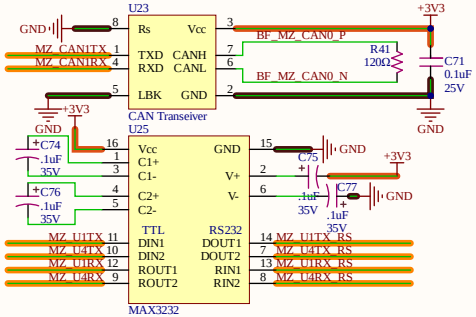




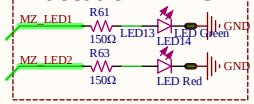
Batteries Reading



MZ Communications



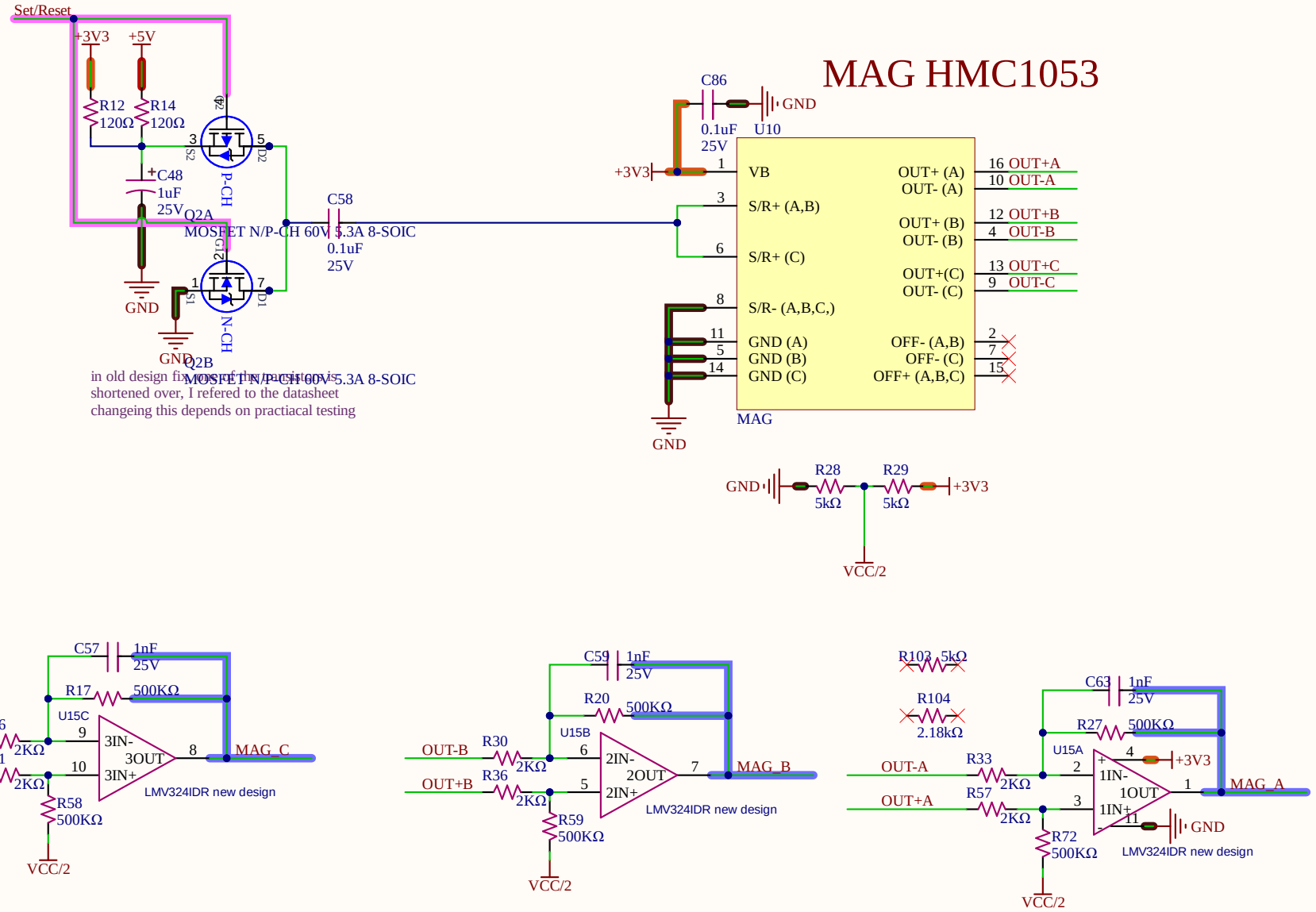
Indecation LEDs



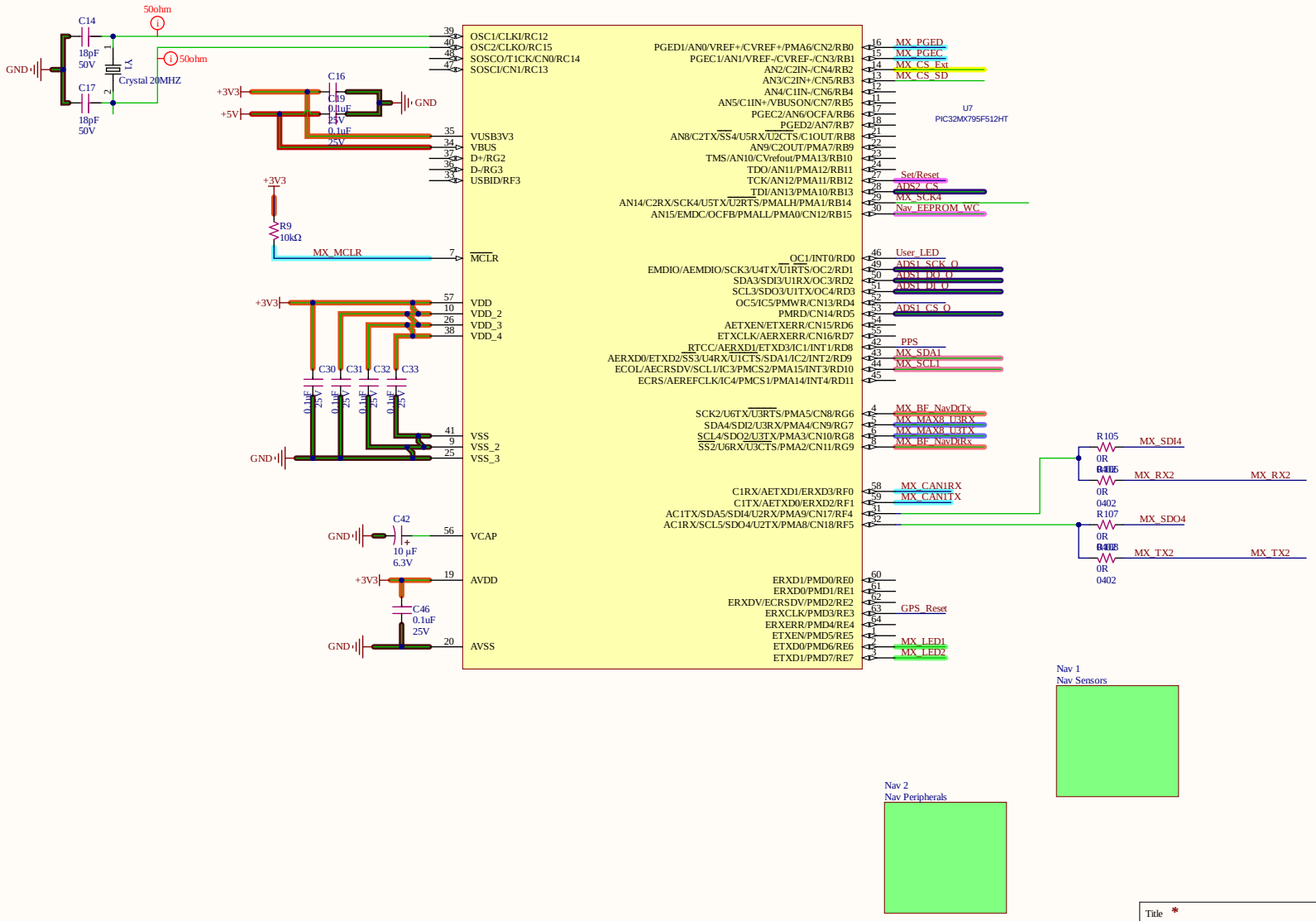
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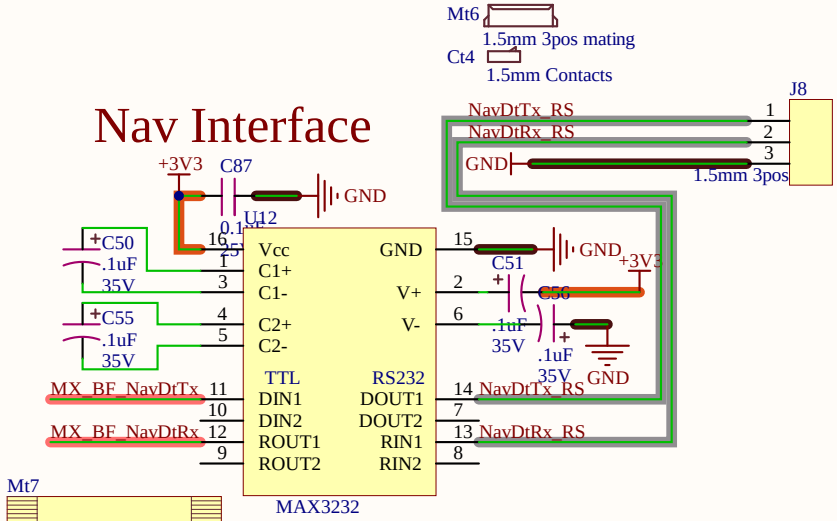
MAG HMC1053

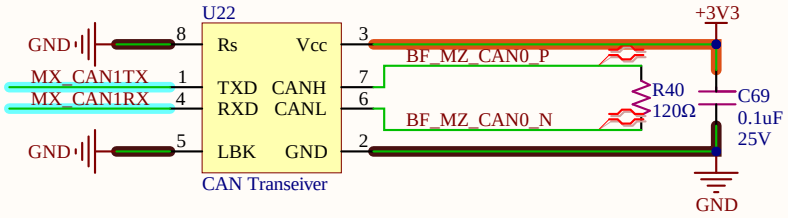


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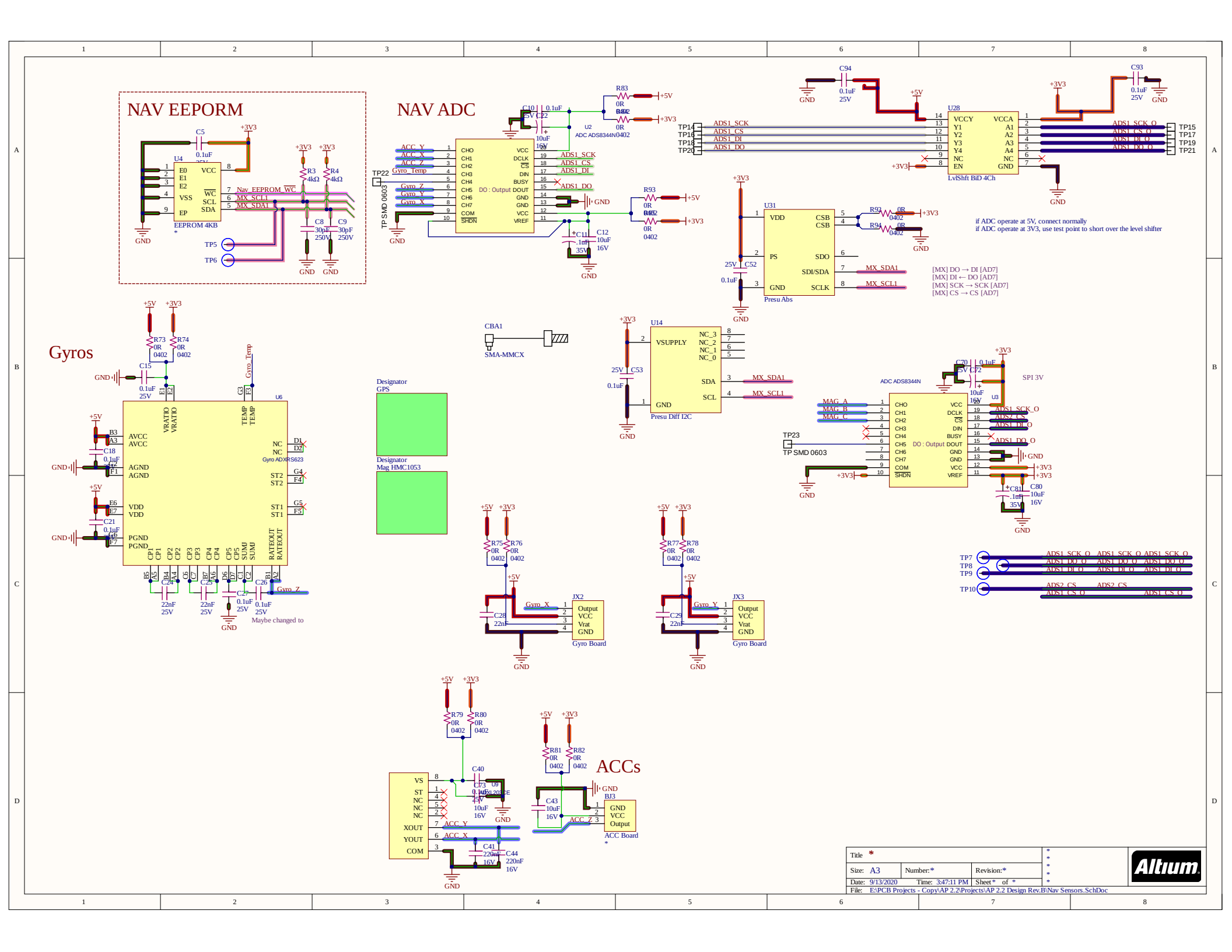


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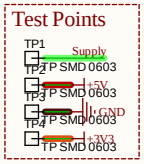


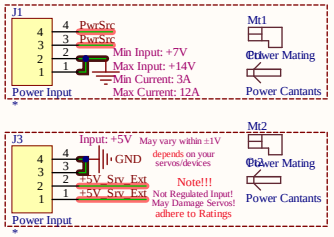


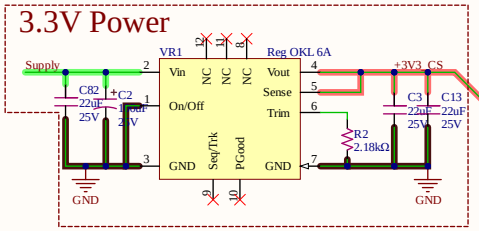
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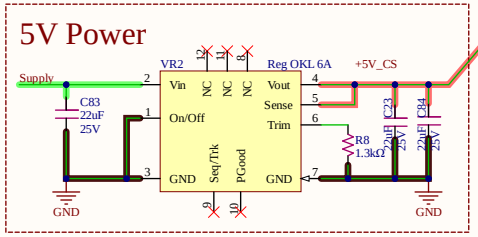

NAV EEPORM
NAV ADC
Gyros
ACCs
EEPROM 4KB
Nav_EEPROM_WC
MX_SCL1
MX_SDA1
ADC ADS8344N
ADS1_SCK
ADS1_CS
ADS1_DI
ADS1_DO
LvlShft BiD 4Ch
Presu Abs
Presu Diff I2C
SMA-MMCX
Designator GPS
Designator Mag HMC1053
Gyro ADXRS623
Gyro Board
ACC Board
SPI 3V
if ADC operate at 5V, connect normally
if ADC operate at 3V3, use test point to short over the level shifter
[MX] DO → DI [AD7]
[MX] DI ← DO [AD7]
[MX] SCK → SCK [AD7]
[MX] CS → CS [AD7]
Maybe changed to
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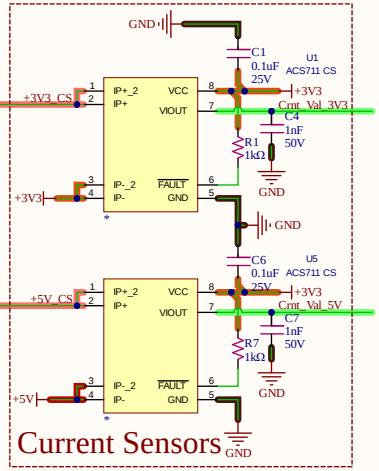
Power System

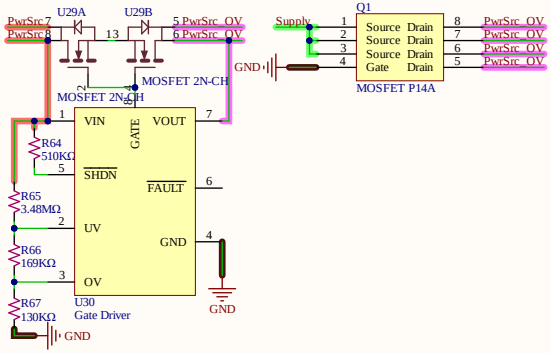


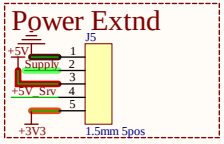


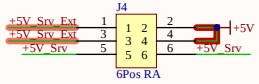








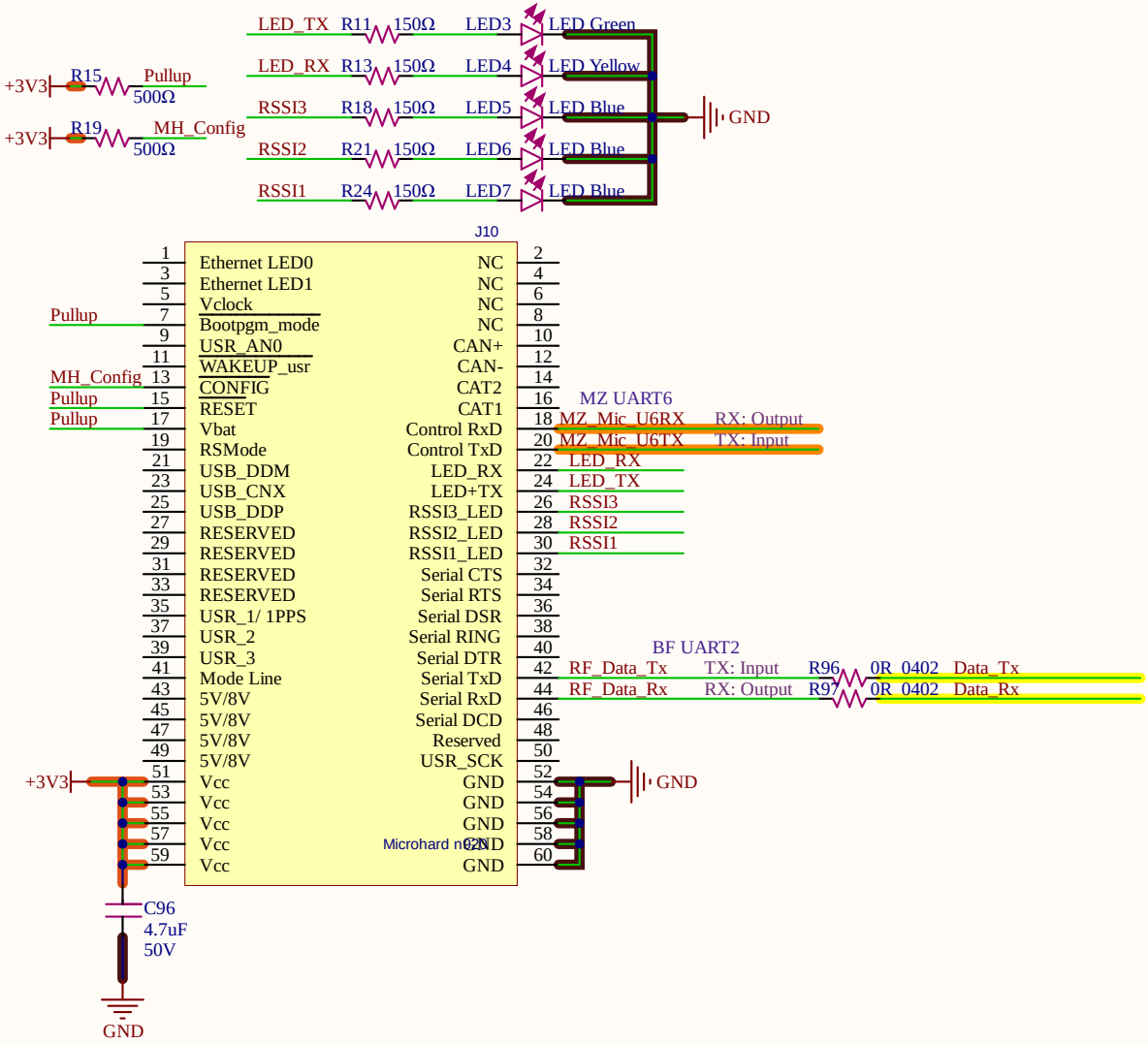




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Microhard Connector



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